

Data Structures and Algorithms - 2025 - 2026

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Status	Finished
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Duration	6 mins 11 secs
Marks	8.00/9.00
Grade	8.89 out of 10.00 (88.89%)

Question 1

Correct

Mark 1.00 out of 1.00

Flag question

The following 4 questions will be related to this example:

Consider an initially empty hash table, with $m = 17$, where coalesced chaining is used for collision resolution and the division method is used for the hash function. In this table the following elements are added in the given order: 13, 24, 35, 30, 58, 170, 92, 66, 38, 47.

How many collisions are in the table? Consider as collision every element which is not on the position to which it hashes.

☐ No collisions

☐ 1 collision

☐ 3 collisions

☒ 6 collisions

☐ 10 collisions

Your answer is correct.

Considering $m = 17$, the value of the hash function for the given elements are:

$h(13) = 13$ $h(24) = 7$ $h(35) = 1$ $h(30) = 13$ $h(58) = 7$ $h(170) = 0$ $h(92) = 7$ $h(66) = 15$ $h(38) = 4$ $h(47) = 13$

Inserting them in this order will give us the following table:

pos	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
elem	30	35	58	170	92	38	47	24						13		66	
next	3	-1	4	6	5	-1	1	2	-1	-1	-1	-1	0	-1	-1	-1	

Collisions are elements: 30, 58, 170, 92, 38, 47.

The correct answer is:
6 collisions

Question 2

Correct

Mark 1.00 out of 1.00

Flag question

Considering the hash table from the previous question.

Assume that you have to add the same elements, but you can change the order in which they are added. In the best case, how much would then be the number of collisions in the table?

Consider collisions as before: the elements which are not on the position where they hash to.

☐ No collisions

☐ 1 collision

☐ 2 collisions

☒ 4 collisions

☐ 6 collisions

Your answer is correct.

We can have collisions in a coalesced chaining hash table in two ways:

- two elements hash to the same position - you cannot avoid these collisions.
- an element hashed to a position which is occupied by another element which hashes to a different position (ex. in case of 170, it hashes to 0, but it could not been added to position 0 because it was occupied by element 13, which hashed to 13, but it was occupied). - these collisions can be avoided / reduced if we first add the elements with different value for the hash function: 13, 24, 35, 170, 66, 38. These can all go to their "normal" position. Then 30, 58, 92 and 47 would be the collisions.

The correct answer is:
4 collisions

Question 3

Correct

Mark 1.00 out of 1.00

Flag question

Consider the hash table with coalesced chaining from the previous question(s). What is the load factor of the table (rounded to 2 decimals)?

Answer: 0.59

Load factor is nr of elements / nr of positions, in our case $10/17 = 0.588$

The correct answer is: 0.59

Question 4

Correct

Mark 1.00 out of 1.00

Flag question

Consider the same hash table with coalesced chaining that was used in the previous question(s). If we performed a search for 64, how many positions from the table would be checked before returning false?

- ☐ 0
- ☐ 1
- ☐ 2
- ☒ 4
- ☐ 6
- ☐ 10
- ☐ 13
- ☐ 17

Your answer is correct.

$h(64) = 13$. So we check the following positions: 13 (not there) -> 0 (not there) -> 3 (not there) -> 6 (not there). Since the next of position 6 is -1, we would stop here.

The correct answer is:
4

Question 5

Correct

Mark 1.00 out of 1.00

Flag question

Consider the following hash table with open addressing, linear probing and the division method for the simple hash function. In what order could the elements have been added in the table?

15	16	10	31	2	73
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- ☐ 10, 16, 31, 15, 73, 2
- ☒ 16, 10, 15, 31, 2, 73
- ☐ 73, 15, 10, 16, 31, 2, 73
- ☒ 10, 16, 31, 2, 73, 15
- ☒ 10, 15, 16, 31, 2, 73
- ☐ 15, 16, 10, 2, 31, 73

Your answer is correct.

The size of the table is $m = 7$, so the hash function used is $h(k,i) = (k \% 7 + i) \% 7$.

There are two ways of solving the problem:

1. Check every possible answer and see if it produces the required table.
2. Try to come up with some rules regarding the order in which elements had to be inserted and check those rules in the solutions. Let's see this version. Let's compute the value of the hahsfunction for $i = 0$ for every element:

$h(15,0) = 1$

$h(16, 0) = 2$

$h(10, 0) = 3$

$h(31, 0) = 3$

$h(2, 0) = 2$

$h(73, 0) = 3$

15 is on its own position, no collision there. So it does not matter when 15 is added.

16 is on its own position, but 2 should also be on position 2. So we know that 16 needs to be added before 2 (otherwise 2 would go on position 2).

10 is on its own position, but 31 and 73 should also be on position 3. So we know that 10 was added before 31 and 73.

31 should be on position 3, but it is on position 4, so we know that position 3 was already occupied when 31 was added.

2 should be on position 2, but it is on position 5. Since in case of linear probing we consider consecutive positions, the fact that 2 is on position 5 means that positions 2, 3 and 4 were occupied when it was added.

Similarly in case of 73.

So we know that 31 had to be inserted after 16

2 had to be inserted after 16 and 31

73 had to be inserted after 16, 31 and 73

The correct answers are:
16, 10, 15, 31, 2, 73,
10, 15, 16, 31, 2, 73,
10, 16, 31, 2, 73, 15

Question 6

Correct

Mark 1.00 out of 1.00

Flag question

Consider the hash table below where open addressing and linear probing was used (with the division method for the simple hash function). XXX denotes a deleted element. Assuming that all elements were added first and then the deletion happened (and after the deletion there were no other operations performed) what value could have been the deleted one?

36	xxx	12		24	43	44
----	-----	----	--	----	----	----

- ☐ 50
- ☒ 19
- ☒ 15
- ☐ 66
- ☐ 74

Your answer is correct.

The table has 9 positions so we know that $m = 9$.

If we look at the possible answers:

$h(19, 0) = 1$, so 19 could have been on position XXX

$h(74, 0) = 2$. Position 2 is currently empty, so if 74 was in the table it had to be on position 2.

$h(50, 0) = 5$. Position 5 is currently empty, so if 50 was in the table, it had to be on position 5

$h(15, 0) = 6$. Position 6 is occupied, and if 15 was added after adding 24, 43, 44 and 36 then it could have been on position XXX

$h(66, 0) = 3$. Position 3 is occupied, but if 66 was on the table it had to be on position 4 (assuming that it was added after 12).

The correct answers are:
19,
15

Question 7

Correct

Mark 1.00 out of 1.00

Flag question

We have a hashtable with $m = 6$ positions. Assuming that we use linear probing and the division method for the simple hash function, what is the probe sequence of element 19? Write all elements of the probe sequence separated by a space.

Answer: 1 2 3 4 5 0

The hash function we will use is: $h(k, i) = (k \% 6 + i) \% 6$. k is 19 and i will go from 0 to 5 (inclusive). So the probe sequence will be: 1 2 3 4 5 0

The correct answer is: 1 2 3 4 5 0

Question 8

Correct

Mark 1.00 out of 1.00

Flag question

We have a hashtable with $m = 6$ positions. Assuming that we use quadratic probing with $c1 = 3$ and $c2 = 2$ and the division method for the simple hash function, what is the probe sequence of element 19? Write all elements of the probe sequence separated by a space.

Answer: 1 0 3 4 3 0

The hash function we will use is: $h(k, i) = (k \% 6 + 3 * i + 2 * i^2) \% 6$. k is 19 and i will go from 0 to 5 (inclusive). So the probe sequence will be: 1 0 3 4 3 0

The correct answer is: 1 0 3 4 3 0

Question 9

Incorrect

Mark 0.00 out of 1.00

Flag question

We have a hashtable with $m = 6$ positions. Assuming that we use double hashing with the division method for the simple hash function and $h^1(k) = 1 + (k \% (m-1))$ as the second hash function, what is the probe sequence of element 19? Write all elements of the probe sequence separated by a space.

Answer: 1 5 3 1 5 3

The hash function we will use is: $h(k, i) = (k \% 6 + i * (1 + k \% 5)) \% 6$. k is 19 and i will go from 0 to 5 (inclusive). So the probe sequence will be: 1 0 5 4 3 2

The correct answer is: 1 0 5 4 3 2

Finish review

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